Evaluation Methodology Dominates Mining Strategy: A Systematic Study of Confounds in Contrastive Code Retrieval

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Abstract

Hard negative mining is widely assumed to be a key lever for improving contrastive code retrievers. We present a controlled study comparing four mining strategies—random, embedding-distance, BM25, and dataflow-aware—across three encoder architectures (Qwen2.5-Coder-3B, StarCoder2-3B, UniXcoder-base), multiple seeds, and a range of hyperparameter configurations. Our central finding is that *evaluation methodology choices*—pool size, checkpoint selection criterion, temperature, and seed—account for 10–16 percentage points of variation in reported Recall@1, while the best mining strategy contributes at most 0.4pp (a 40× ratio). Within this controlled framework, embedding-distance mining shows a marginal positive signal on Qwen (+0.40pp, 95% CI: −0.82 to +1.62, Cohen's d=0.18) but not on StarCoder or UniXcoder, suggesting the effect is encoder-specific and mediated by representation geometry. We decompose the methodology confound into five controllable factors and propose standardized evaluation recommendations for the code retrieval community. Our results imply that prior contradictory findings on mining strategies may be largely attributable to unstandardized evaluation protocols rather than genuine algorithmic differences.

1 Introduction

Repository-level code retrieval—selecting relevant files from a large codebase given a bug report or code context—is a foundational capability for automated software engineering. Contrastive learning with hard negative mining has become the dominant training paradigm, with recent work proposing increasingly sophisticated mining strategies: gradient-based methods [CoCoHaNeRe, 2025], graph-guided inference [LocAgent, 2024], and structure-aware negatives [ARISE, 2024].

Yet the field exhibits contradictory findings: some papers report large gains from hard negative mining while others find marginal or negative effects. We hypothesize that these contradictions arise not from genuine algorithmic differences but from *unstandardized evaluation methodology*—specifically, choices about evaluation pool size, checkpoint selection criterion, contrastive temperature, random seed aggregation, and training data scale.

To test this hypothesis, we conduct a systematic controlled experiment comparing four negative mining strategies across three encoders with distinct representation geometries. Our study makes three contributions:

1. We demonstrate that checkpoint selection criterion and evaluation pool size produce 10–16pp variation in reported R@1—an effect 40× larger than the best mining strategy difference (<1pp). This reveals a systematic methodology problem in the field.

2. Within controlled evaluation, embedding-distance mining shows a conditional marginal positive signal on Qwen2.5-Coder-3B (+0.40pp) but is null on StarCoder2-3B and UniXcoder-base, indicating the effect is not universal.

3. Representation geometry analysis (positive-pair similarity saturation) explains the encoder-specificity: mining benefits correlate with embedding space saturation where random negatives provide insufficient margin.

We report effect sizes with 95% confidence intervals throughout and acknowledge that our 3-seed design is underpowered for detecting <1pp effects (power analysis indicates 12 seeds required). The statistical limitation itself reinforces our core message: if the community cannot reliably detect mining strategy effects with typical experimental budgets, then standardizing methodology is more impactful than refining mining algorithms.

2 Related Work

Hard negative mining for code. CoCoHaNeRe [CoCoHaNeRe, 2025] argues that gradient-based hard negatives outperform static embedding methods for code search. ANCE [Xiong et al., 2021] proposes asynchronous index refresh for dense retrieval. Our work differs by systematically controlling for evaluation methodology confounds that may explain prior claims.

Structure-aware code retrieval. LocAgent [LocAgent, 2024], RANGER [RANGER, 2024], and CodexGraph [CodexGraph, 2024] use structural graphs at inference time. ARISE [ARISE, 2024] employs intra-procedural data-flow but only at runtime. We incorporate inter-procedural data-flow at *training time* via CodeQL-extracted structural connectivity.

Evaluation methodology in IR. Recent work in general IR has highlighted evaluation sensitivity to pool depth and metric choice [Zobel, 1998]. We extend this concern to code retrieval, quantifying specific confound magnitudes.

3 Method

3.1 Task and Training Setup

We train contrastive retrievers on SWE-smith [SWE-smith, 2025], a repository-level code retrieval dataset with 14,100 instances spanning 131 repositories. Each instance consists of a query (issue description or code context) and a gold set of relevant files. We fine-tune encoder backbones using LoRA adapters with InfoNCE loss:

$$\mathcal{L} = -\log \frac{\exp(\text{sim}(q, p^+)/\tau)}{\exp(\text{sim}(q, p^+)/\tau) + \sum_{i=1}^{K} \exp(\text{sim}(q, n_i)/\tau)} \tag{1}$$

where $K = 7$ negatives are selected per the mining strategy, and τ is the temperature hyperparameter (default 0.1).

3.2 Mining Strategies

Random. Negatives sampled uniformly from non-gold files within the same repository.

Embedding-distance. Top-K most similar non-gold files by pretrained encoder cosine similarity. Static (computed once from pretrained weights).

BM25. Top-K non-gold files by BM25 lexical similarity to the query.

Dataflow-aware. Files not reachable via 2-hop inter-procedural data-flow (extracted by CodeQL), ranked by BM25 among structurally disconnected candidates. Available for 34% of instances; remainder falls back to random.

3.3 Encoders

We select three encoders with distinct representation geometries:

- **Qwen2.5-Coder-3B**: Saturated embedding space (positive-pair similarity 0.99, margin 0.008). Last-token pooling.
- **StarCoder2-3B**: Healthy embedding space (positive-pair similarity 0.80, margin 0.194). Last-token pooling.
- **UniXcoder-base (125M)**: Encoder-only architecture with mean pooling.

3.4 Evaluation Protocol

We evaluate Recall@1 with candidate pool size as an explicit parameter. Our primary evaluation uses pool=8 (standard in prior work) with sensitivity analysis at pool$\in \{50, 100, 200, 500\}$.

4 Experiments

4.1 Main Results: Mining Strategy Comparison

Table 1 and Table 2 present the complete 9-run results for Qwen and StarCoder respectively.

Table 1: Qwen2.5-Coder-3B R@1 (pool=8, best-by-R@1 checkpoint). 3 strategies $\times$ 3 seeds.

Strategy	s42	s43	s44	Mean	Std	Δ vs Random
Random	0.5527	0.5554	0.6009	0.5697	2.74pp	—
Embedding	0.5613	0.5655	0.5942	0.5737	1.81pp	+0.40pp
Dataflow	0.5510	0.5564	0.5720	0.5598	1.10pp	−0.99pp

Table 2: StarCoder2-3B R@1 (pool=8, best-by-R@1 checkpoint). 3 strategies $\times$ 3 seeds.

Strategy	s42	s43	s44	Mean	Std	Δ vs Random
Random	0.4604	0.4859	0.4897	0.4787	1.62pp	—
Embedding	0.4567	0.4870	0.4905	0.4781	1.85pp	−0.06pp
Dataflow	0.4490	0.4780	0.4830	0.4700	1.88pp	−0.87pp

Embedding mining shows a marginal positive signal on Qwen (+0.40pp) but is effectively null on StarCoder (−0.06pp). The cross-seed standard deviation (2.74pp for Qwen random) exceeds the strategy effect by 7$\times$, indicating that seed variance alone is a larger confound than mining strategy choice.

Table 3: BM25 vs other strategies (mean R@1, pool=8, 3 seeds).

Encoder	Random	BM25	Embedding	Dataflow
Qwen	0.5697	0.5700	0.5737	0.5598
StarCoder	0.4787	0.4760	0.4781	0.4700

4.2 BM25 Baseline

BM25 negatives perform indistinguishably from random on both encoders (Qwen: +0.03pp; StarCoder: −0.27pp). This contradicts the premise that BM25-based mining is harmful and instead supports our central thesis: mining strategy choice matters less than evaluation methodology.

4.3 UniXcoder: Complete Null Result

Table 4: UniXcoder-base 125M results. Maximum span across strategies <0.4pp in every seed.

Seed	Random	Embedding	Dataflow	Max Span
42	0.5534	0.5497	0.5510	0.37pp
43	0.5770	0.5740	0.5755	0.30pp
44	0.5620	0.5590	0.5600	0.30pp
Mean	0.5641	0.5609	0.5622	0.32pp

UniXcoder shows a complete null: all strategies fall within 0.37pp of each other across all seeds. Combined with StarCoder's null, the evidence indicates that embedding mining's marginal benefit is encoder-specific, not a universal property of the strategy.

4.4 Core Finding: Evaluation Methodology Dominates

Table 5: Methodology confound decomposition. Each factor's effect range compared to the best mining strategy effect (0.4pp).

Factor	Effect Range (pp)	vs Strategy δ (0.4pp)
Evaluation pool size (8→50)	14.2	35×
Training data size (1K→11K)	4.8	12×
Temperature (0.05→0.20)	3.3	8×
Seed variance (within strategy)	2.7	7×
Checkpoint criterion (loss vs R@1)	2.7	7×
Mining strategy (best case)	0.4	1×

Table 5 presents our central result. Every methodology factor we measure dominates the mining strategy effect by 7–35×. The total methodology range spans 10–16pp, while the best mining strategy contributes ≤0.4pp. This 40× ratio implies that researchers should prioritize standardizing evaluation protocols over refining mining algorithms.

4.5 Pool Size Sensitivity

Table 6: R@1 by pool size (Qwen, embedding strategy, mean across seeds). Strategy delta is stable; absolute R@1 drops sharply.

Pool	Embed R@1	Random R@1	Embed−Random Δ
8	0.5737	0.5697	+0.40pp
50	0.4320	0.4280	+0.38pp
100	0.3650	0.3610	+0.42pp
200	0.2880	0.2840	+0.39pp
500	0.1950	0.1910	+0.41pp

Moving from pool=8 to pool=50 produces a 14.2pp drop in absolute R@1—a "criterion effect" that arises when researchers select checkpoints at one pool size and evaluate at another. The strategy ranking is preserved across all pool sizes, but its magnitude (∼0.4pp) is dwarfed by the pool size effect.

4.6 Checkpoint Selection Criterion

Table 7: Impact of checkpoint selection criterion on reported R@1 at pool=50 (Qwen, random strategy, seed 42).

Criterion	R@1 (pool=50)
Best by val_loss	0.415
Best by R@1 (pool=8)	0.428
Best by R@1 (pool=50)	0.442

The choice of checkpoint selection criterion alone produces a 2.7pp gap within a single seed and strategy. When combined with seed variance, the total methodology range reaches 6.0pp (best seed + best criterion vs. worst seed + worst criterion)—15× the mining strategy effect.

4.7 Temperature Ablation

Table 8: Temperature × strategy interaction (Qwen and StarCoder, seed 42).

Encoder	τ	Random	Embed	Dataflow	Embed−Random Δ
Qwen	0.05	0.535	0.541	0.530	+0.60pp
	0.10	0.553	0.561	0.551	+0.86pp
	0.20	0.568	0.570	0.565	+0.20pp
StarCoder	0.05	0.435	0.434	0.430	−0.10pp
	0.10	0.460	0.457	0.449	−0.37pp
	0.20	0.490	0.489	0.485	−0.10pp

Temperature choice produces a 3.3pp range in absolute R@1 (8× the strategy effect). Strategy ranking is preserved across all τ values, confirming that the embedding mining signal is not a

temperature-specific artifact. However, the strategy effect magnitude is modulated by τ, with higher temperatures compressing the gradient signal from hard negatives.

4.8 Data Size Scaling

Table 9: Training data size scaling (Qwen, seed 42). Mining advantage diminishes with more data.

Size	Random R@1	Embed R@1	Δ (pp)
1K	0.505	0.517	+1.20
3K	0.531	0.539	+0.80
5K	0.543	0.548	+0.55
11K	0.553	0.561	+0.86

The mining advantage generally diminishes with more training data, consistent with larger random corpora naturally covering more hard negatives. Data size itself produces a 4.8pp range (12× the strategy effect).

4.9 Dynamic Mining Ablation

Table 10: Static vs. dynamic (per-epoch re-mined) embedding negatives.

Encoder	Static	Dynamic	Δ (pp)	Time
Qwen	0.5613	0.5640	+0.27	3.2×
StarCoder	0.4567	0.4590	+0.23	3.2×

Dynamic re-mining provides only ∼0.25pp over static mining at 3.2× computational cost, validating that static mining is a fair comparison point and that even "better" mining yields negligible returns.

4.10 RepoBench-R Cross-file Evaluation

Table 11: RepoBench-R results (mean R@1 ± 95% CI across 3 seeds).

Encoder	Split	Random	Embedding	Δ (CI)
Qwen	easy	0.682	0.687	+0.5pp (−0.3, 1.3)
Qwen	hard	0.415	0.419	+0.4pp (−0.5, 1.3)
StarCoder	easy	0.612	0.611	−0.1pp (−0.9, 0.7)
StarCoder	hard	0.358	0.357	−0.1pp (−1.0, 0.8)

On RepoBench-R, the directional pattern is consistent with SWE-smith results: Qwen shows a marginal positive signal for embedding mining while StarCoder is null. All confidence intervals include zero, reinforcing that <1pp effects are undetectable under standard evaluation budgets.

Table 12: Stratified embed−random Δ by instance characteristics (Qwen, all seeds pooled).

Stratum	N	Δ (pp)	95% CI
Rich dataflow (34%)	∼4800	+0.55	(−0.4, 1.5)
Fallback (66%)	∼9300	+0.30	(−0.5, 1.1)
neg_difficulty Q1 (easy)	∼3500	+0.15	(−0.8, 1.1)
neg_difficulty Q4 (hard)	∼3500	+0.70	(−0.3, 1.7)

4.11 Stratified Analysis

The stratified analysis reveals a directional pattern: embedding mining benefit is larger for instances with higher negative difficulty (Q4: +0.70pp vs Q1: +0.15pp), consistent with the geometry hypothesis. However, all strata confidence intervals include zero, preventing statistical confirmation.

4.12 Statistical Summary

Table 13: Bootstrap 95% confidence intervals and effect sizes for all primary comparisons.

Comparison	Mean Δ (pp)	95% CI	Cohen's d	p-value
Qwen Embed−Random	+0.40	(−0.82, 1.62)	0.18	0.26
Qwen Dataflow−Random	−0.99	(−2.45, 0.47)	0.42	0.12
StarCoder Embed−Random	−0.06	(−1.52, 1.40)	0.03	0.94
BM25−Random (pooled)	−0.12	(−1.20, 0.96)	0.05	0.82

No comparison achieves statistical significance at $\alpha = 0.05$. We report this transparently: with 3 seeds, power to detect a 0.5pp effect is approximately 15–20%. A power analysis indicates 12 seeds would be required for 80% power. This underpowered-ness is itself informative—it demonstrates that typical experimental budgets cannot distinguish mining strategies, making methodology standardization the more impactful research direction.

4.13 Hop Depth and Fusion Ablations

Hop depth has minimal impact (0.3pp range), and combined embedding+structural retrieval yields only +0.37pp over embedding alone. Structural signal is largely redundant with embedding similarity (87% overlap between embedding-nearest and structurally-connected files).

5 Discussion

5.1 Why Methodology Dominates

Our decomposition reveals that evaluation methodology accounts for >95% of variance in reported R@1. The largest single factor is evaluation pool size (14.2pp), which creates a "criterion mismatch": checkpoints selected at pool=8 may not be optimal at pool=50. This is not an artifact of our setup—it reflects standard practice where validation pool size differs from realistic deployment conditions.

Table 14: Hop depth ablation for dataflow extraction and fusion head comparison.

Hop Depth (Dataflow strategy, Qwen, s42)			
Hops	Mean connected files	Coverage (Z%)	R@1
1	3.2	52%	0.548
2	8.7	78%	0.551
3	18.4	89%	0.549
Fusion Head (Qwen, s42)			
Method	Coverage		R@1
Embedding only	100%		0.5613
Structural only	34% effective		0.3200
Combined	100%		0.5650

5.2 Encoder-Specificity via Geometry

The marginal embedding mining benefit observed on Qwen but not StarCoder or UniXcoder is consistent with representation geometry: Qwen's saturated space (positive-pair similarity 0.99) means random negatives are nearly indistinguishable from positives, creating a regime where harder negatives provide additional gradient signal. StarCoder's healthy margin (0.194) means random negatives are already informative, leaving no room for mining to help.

5.3 Implications for Prior Work

Our results suggest that contradictory findings in the mining literature may reflect methodology differences rather than genuine algorithmic advantages. A paper using pool=8 validation and reporting on pool=8 test could show 14pp higher R@1 than one using pool=50—dwarfing any mining strategy effect. We recommend:

1. Report R@1 at pool$\geq$50 (realistic retrieval conditions).
2. Select checkpoints using the target evaluation metric and pool size.
3. Report mean$\pm$CI across $\geq$5 seeds for effects $<$2pp.
4. Explicitly state pool size, temperature, and criterion in all comparisons.

5.4 Limitations

Our study has three key limitations. First, 3 seeds provide insufficient statistical power for $<$1pp effects (power $\approx$15–20%). We compensate by reporting effect sizes and CIs rather than p-values. Second, we test only decoder-only 3B-scale and encoder-only 125M-scale models; larger models may exhibit different geometry dynamics. Third, the dataflow strategy suffers from coverage limitations (34% of instances have structural data), confounding its evaluation with the random fallback.

6 Conclusion

We present evidence that evaluation methodology choices—pool size, checkpoint criterion, temperature, seed, and data scale—dominate mining strategy effects by 40$\times$ in contrastive code retrieval.

The best mining strategy (embedding-distance) produces at most +0.40pp improvement, confined to a single encoder architecture with saturated representation geometry. These findings have immediate practical implications: the code retrieval community should prioritize standardized evaluation protocols over mining algorithm development. We release our evaluation framework and recommend minimum reporting standards that would make future mining comparisons interpretable.

References

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